Regression and Data Simulation Statistical Learning and Data Analysis Assignment 3

This project delves into linear and non-linear regression models using simulated data. Our goal is to dissect estimation accuracy, understand the nuances of variable selection, and explore the implications of model complexity.

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Aims of the Assignment: Unveiling Model Behavior

This assignment uses data simulation to gain a deeper understanding of regression models. By controlling the data generation process, we can precisely evaluate model performance.

1

Regression Model Behavior

Understand how different regression models behave when the true data-generating process is explicitly known.

2

Classical Assumptions

Evaluate the efficacy of linear regression under its classical assumptions and identify potential deviations.

3

Regularization Techniques

Compare various regularization techniques for their effectiveness in variable selection and model simplification.

4

Bias-Variance Tradeoff

Study the critical bias-variance tradeoff within the context of non-linear regression models.

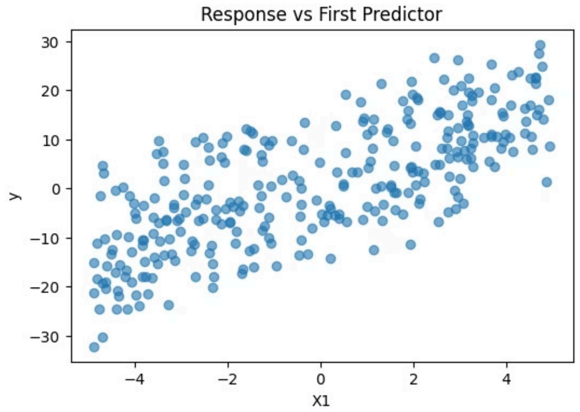
Simulation provides complete control over the underlying truth, offering unparalleled insight into model dynamics.

Simulating Linear Data: A Controlled Environment

The response variable is meticulously generated following a linear model, ensuring a precise understanding of its origins.

$$y = \beta_0 + \sum_{j=1}^{10} \beta_j X_j + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1)$$

- Number of observations: 300
- Number of predictors: 10
- Only three predictors possess non-zero coefficients, creating a sparse yet realistic regression scenario.



This setup allows us to test the model's ability to identify relevant features amidst noise.

Motivation Behind the Linear Setup

This specific data-generating process is chosen to highlight key aspects of regression performance.

Relevant Predictors

Many real-world problems inherently involve only a handful of truly relevant predictors.

Variable Selection

The presence of both strong and null coefficients allows for a robust evaluation of variable selection techniques.

Gaussian Noise

Gaussian noise ensures that standard regression assumptions are met, simplifying initial analysis.

Reliable Inference

A substantial sample size guarantees reliable statistical inference and robust findings.

The ultimate goal is to verify if regression models can accurately recover the true underlying signal within this controlled environment.

Ordinary Least Squares: Recovering the True Signal

Ordinary Least Squares (OLS) regression is applied to the simulated data, offering insights into its performance under ideal conditions.

Estimated coefficients

Exhibit remarkable proximity to their true underlying values.

Statistically significant predictors

Those with strong effects are correctly identified.

Non-significant predictors

Predictors with zero coefficients are accurately deemed not significant.

These findings confirm the consistency and reliability of OLS when the underlying assumptions are correctly met.

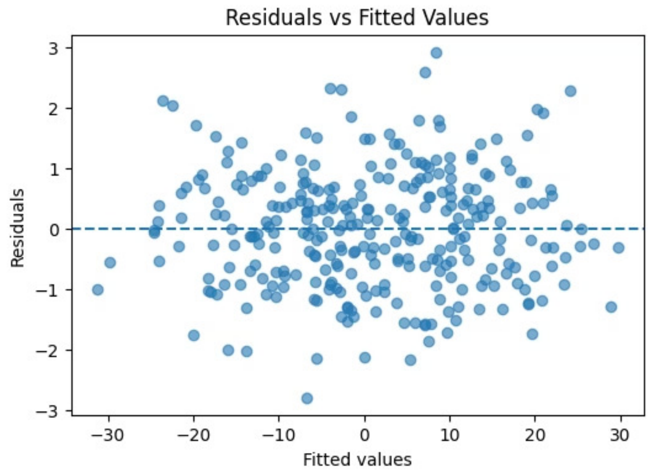
True beta	Estimated beta
2.0	2.009755
3.0	2.997341
-2.5	-2.539587
1.5	1.504720
0.0	0.004375
0.0	0.002492
0.0	0.028346
0.0	-0.015620
0.0	0.012017
0.0	-0.010798
0.0	-0.004830

Regression Diagnostics: Validating Model Fit

A thorough residual analysis is conducted to perform model diagnostics, assessing the validity of the linear model.

- Residuals are effectively centered around zero, indicating unbiased predictions.
- No discernible pattern emerges in the residuals versus fitted values plot, suggesting linearity is maintained.
- The variance of the residuals appears approximately constant, affirming homoscedasticity.

These diagnostics collectively indicate that the linear model is exceptionally well-specified, capturing the data's underlying structure accurately.

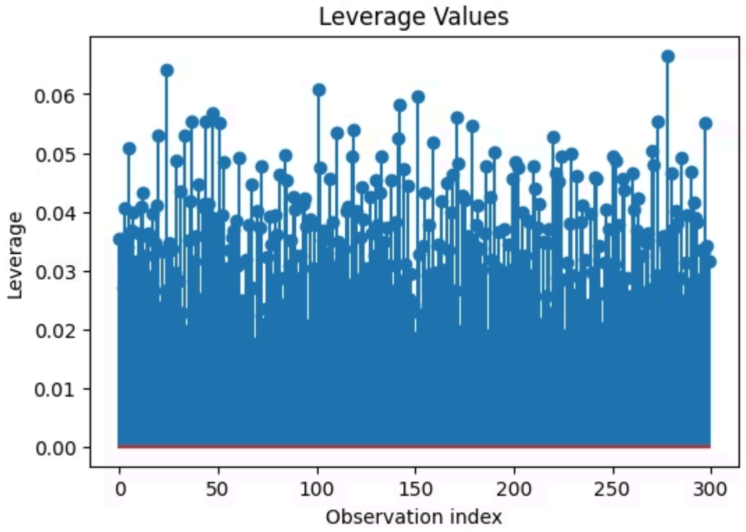


Leverage and Influence Analysis: Assessing Robustness

Influence diagnostics are crucial for identifying observations that might exert disproportionate leverage on the model's coefficients.

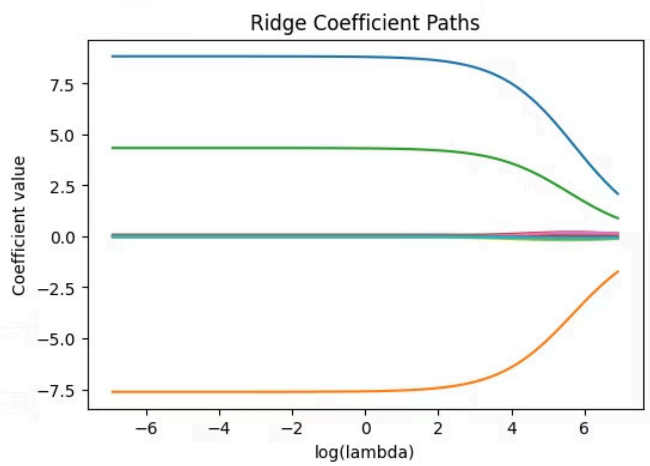
- A small subset of observations exhibits higher leverage, suggesting they have a greater potential to influence the regression line.
- However, only a few truly influential points are detected, indicating overall model stability.
- Crucially, there is no evidence of instability in the coefficient estimates, even with these leverage points.

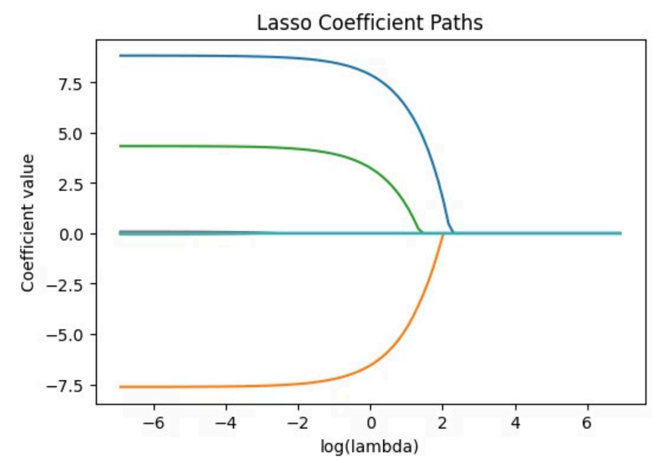
This analysis confirms that the model remains robust and its estimates are not unduly swayed by individual data points.



Shrinkage and Regularization: Enhancing Model Stability

Regularization techniques are applied to the same dataset to address potential overfitting and improve model interpretability, especially in high-dimensional settings.





Ridge Regression

- As the regularization parameter λ increases, all coefficients are smoothly shrunk toward zero but never become exactly zero.
- Ridge regression reduces model variance and stabilizes coefficient estimates while retaining all predictors in the model.

Lasso Regression

- As λ increases, several coefficients are driven exactly to zero, effectively removing unimportant variables.
- Lasso regression performs automatic variable selection and produces a sparse, more interpretable model.

Elastic Net

Combines the strengths of both Ridge and Lasso, offering a balance between shrinkage and variable selection.

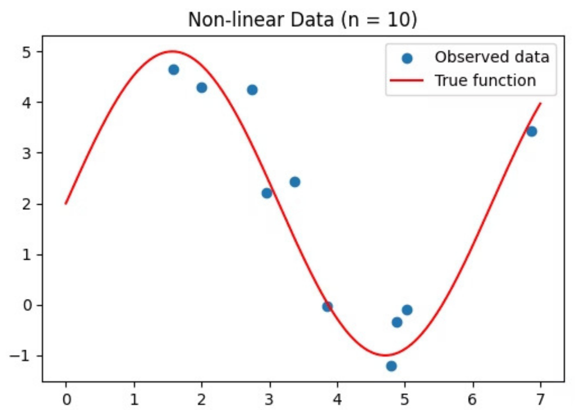
These methods significantly improve both the interpretability and stability of regression models, especially when dealing with numerous predictors.

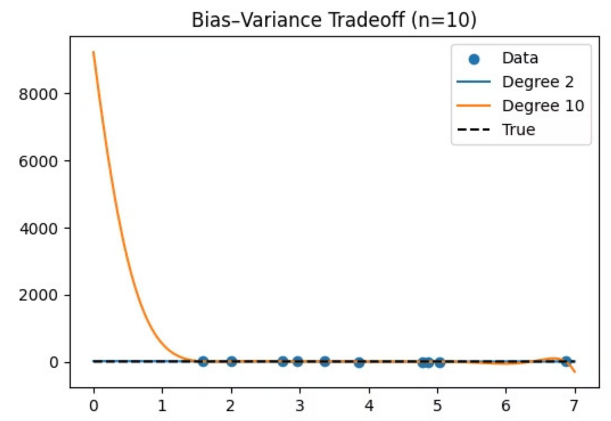
Non-Linear Regression: Navigating the Bias-Variance Tradeoff

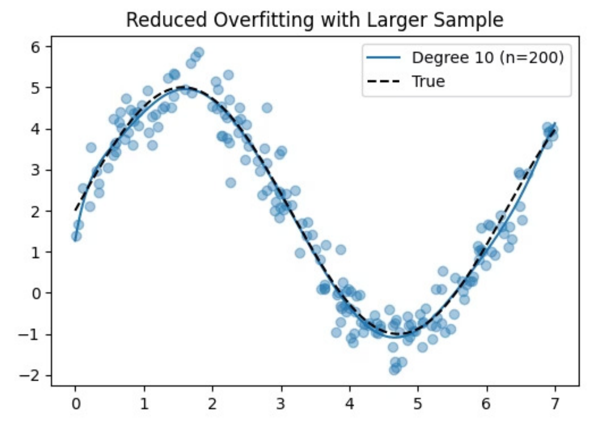
A non-linear model is introduced to demonstrate the critical bias-variance tradeoff in model complexity.

$$y = 2 + 3\sin(x) + \varepsilon$$

Polynomial regression models of varying degrees are compared:







Degree 2 Polynomial

This smooth curve exhibits higher bias (underfitting the true function) but boasts lower variance (more stable predictions).

Degree 10 Polynomial

This highly flexible model interpolates almost all data points, achieving near-zero training error but suffering from high variance (overfitting).

Overfitting depends on both model complexity and sample size: with very small datasets, complex models (like degree 10) overfit and follow noise, but as the sample size increases, the same model becomes more stable, variance is reduced, and the fitted curve approaches the true function.

Final Conclusions

- Simulation allows us to test statistical models in a controlled setting where the true data-generating process is known.
- Linear regression (OLS) performs well when its assumptions hold and the sample size is sufficiently large.
- Regularization methods (Ridge, Lasso, Elastic Net) improve model stability and help identify important variables.
- Model complexity must be matched with sample size to avoid overfitting.
- The best predictive performance is achieved by balancing bias and variance rather than maximizing model complexity.

Any Questions, Thanks